

COMPUTER ARCHITECTURE – MST QUESTION BANK

3 Marks, 5 Marks and 7 Marks Questions

SECTION A – 3 MARKS QUESTIONS (20)

1. Define Register Transfer Language (RTL). Explain any two RTL statements with examples.
2. Differentiate between register transfer, arithmetic micro-operation, and logic micro-operation with one example each.
3. A 4-bit register contains 1011. Find the result after: (a) Logical left shift (b) Logical right shift (c) Arithmetic right shift.
4. Perform 10110110 AND 01101101 and 10110110 XOR 01101101. Show the results.
5. Explain the functions of AR, PC, IR, DR, and AC in a basic computer.
6. Explain the three major phases of an instruction cycle.
7. Explain memory-reference instructions with any three suitable examples.
8. Differentiate between hardware interrupt and software interrupt.
9. Differentiate between machine language and assembly language.
10. What is a subroutine? Explain its two major advantages in assembly language programming.
11. What is control memory? Why is it required in a microprogrammed control unit?
12. Explain the functions of CAR and control memory in a microprogrammed control unit.
13. Differentiate between hardwired and microprogrammed control in any three points.
14. Define effective address. Find the effective address if $R1 = 500$ and the displacement is 250 in indexed addressing.
15. Explain immediate, direct and indirect addressing modes with examples.
16. Differentiate between register stack and memory stack.
17. Explain zero-address and one-address instruction formats with examples.
18. State any three important differences between RISC and CISC architectures.

SECTION B – 5 MARKS QUESTIONS (20)

19. Explain Register Transfer Language and common bus system with a suitable diagram and RTL examples.
20. Explain arithmetic, logic and shift micro-operations with suitable examples. Differentiate between logical and arithmetic right shifts.
21. Given $R1 = 10110101$ and $R2 = 00111011$, perform: (a) $R1 + R2$ (b) $R1 - R2$ using 2's complement (c) $R1 \text{ AND } R2$ (d) $R1 \text{ XOR } R2$.
22. Explain the organization and working of an Arithmetic Logic Unit (ALU) with a neat block diagram.
23. Explain the complete fetch and decode cycle of a basic computer

24. Explain the functions of the major registers in a basic computer and describe how they participate during common bus system.
25. Explain the following memory-reference instructions with suitable examples: BSA, NBR, LDA, STA, BUN.
26. Explain the interrupt cycle of a basic computer. Why is an interrupt mechanism necessary?
27. Explain the concept of subroutine and describe how a subroutine can be called and returned from.
28. Explain the organization of a microprogrammed control unit with a neat block diagram.
29. Explain address sequencing in a microprogrammed control unit and discuss different methods of generating the next microinstruction address.
30. Explain General Register Organization with a suitable block diagram. Show how $R3 \leftarrow R1 + R2$ can be performed.
31. Explain stack organization and the operations PUSH and POP with a suitable example.
32. Explain immediate, direct, indirect, register, register-indirect and indexed addressing modes with examples.
33. Given PC = 1200 and displacement = +350, calculate the effective address using relative addressing. Also explain when relative addressing is useful.
34. Compare RISC and CISC based on instruction set, instruction format, addressing modes, memory access, control unit and execution.

SECTION C – 7 MARKS QUESTIONS (20)

35. Given $R1 = 10101101$ and $R2 = 01010111$, perform: (a) $R3 \leftarrow R1 + R2$ (b) $R4 \leftarrow R1 - R2$ using 2's complement (c) $R5 \leftarrow R1 \text{ AND } R2$ (d) Logical left shift of $R1$ by 2 positions (e) Arithmetic right shift of $R2$ by 2 positions. Identify the type of micro-operation used in each case.
36. Explain the organization of a common bus system using multiplexers. Show how data can be transferred between multiple registers through the common bus and explain the required selection signals.
37. Explain the complete organization of a basic computer with a neat block diagram. Discuss the functions of Memory, ALU, AC, PC, IR, AR, DR, Control Unit and I/O registers.
38. Explain the working of AND, ADD, LDA, STA, BUN, BSA and ISZ instructions of a basic computer. Show suitable RTL operations for any four instructions.
39. Explain the interrupt mechanism and interrupt cycle of a basic computer with a suitable flow diagram. Discuss what happens to the current program when an interrupt occurs.
40. Write an assembly language program/algorithm to add N numbers stored in consecutive memory locations. Explain initialization, memory addressing, loop counter, accumulation, loop termination and result storage.
41. Explain address sequencing in microprogrammed control. Describe how the next microinstruction address can be obtained through: (1) Sequential increment (2) Branching (3) Opcode mapping (4) Subroutine/return mechanism.

42. Explain General Register Organization with a suitable diagram. Show the sequence of control operations required to execute $R3 \leftarrow R1 + R2$. Clearly identify source registers, destination register, bus and ALU operations.
43. Explain stack organization with a neat diagram. Consider: PUSH A, PUSH B, PUSH C, POP, PUSH D, POP. Show the stack contents after each operation and explain the role of the Stack Pointer.
44. Explain zero-address, one-address, two-address and three-address instruction formats with suitable examples. Then explain how $X = (A + B) \times (C - D)$ could be evaluated using different instruction formats.
45. Explain stack organization in detail. Compare register stack and memory stack based on organization, speed, capacity, cost and implementation.
46. Explain the storage program organization in detail. Explain opcode, instruction , address and memory with diagram

Read question carefully .

draw diagram wherever required don't think that if it is not mentioned in question why should I draw. It will fetch you more marks.

Don't overwrite in MCQ ,overwrite will be consider as 0

D2D student mention your class on the top of your answer sheet.

Trust your hard work and preparation because success is built on the effort you put in today.